

Feature Extraction and Classification of EEG for Imaging Left-right Hands Movement

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Abstract—brain-computer interface (BCI) is a system that allows its users to control external devices with brain activity. This paper presents a new method for classifying the off-line experimental electroencephalogram (EEG) signals from the BCI Competition 2003, which achieved higher accuracy. The method has three main steps. First, wavelet coefficient was reconstructed by using wavelet transform in order to extract feature of EEG for mental tasks. At the same time, in frequency extraction, we use the AR model power spectral density as the frequency feature. Second, we combine the power spectral density feature and the wavelet coefficient feature as the final feature vector. Finally, linear algorithm is introduced to classify the feature vector based on iteration to obtain weight of the vector's components. The classified result shows that the effect using feature vector is better than just using one feature. This research provides a new idea for the identification of motor imagery tasks and establishes a substantial theory and experimental support for BCI application. .

Keywords—brain computer interface; EEG; motor imagery; feature extraction; power spectral density; wavelet transform.

I. INTRODUCTION

Interest in developing a new method of the brain-computer interface (BCI) has grown steadily over the past few decades. BCIs create a new communication channel between the brain and an output device by bypassing conventional motor output pathways of nerves and muscle. BCI technology could therefore provide a new communication and control option for individuals who can't otherwise express their wishes to the outside world [1]. What's more, the BCI technology has possible applications in other fields, such as special work over and military affairs and it will be a new way for exchange or control of information and entertainment [2].

Signal processing and classification methods are essential tools in the development of improved BCI technology. Various methods have been used and some have satisfactory accuracy, but at the same time unexpected limitations are also existed, such as depending closely on many unrelated factors, needing to process a large amount of data, difficult to maintain and update with complicated algorithm. After

studying and comparing several processing methods of mental EEG signal, we promote a new method based on the following theory. First of all, wavelet, which is called a mathematical microscope for analyzing signals, has the ability to indicate signal which is localized in time domain or frequency domain. Wavelet transform is an effective mathematical analysis method to extract the feature of signal. By means of wavelet transform, we can obtain the coefficient feature of the time domain needed in our research. The coefficient extracted in the transform can well reflect the feature of the original signal, and there's a considerable advantage that data can be compressed at a large extent. Secondly, the power of Mu rhythm, at 8-12 frequency bands and closely related to human sensorimotor function, are quite different according to left-right hands imagery motor. Based on this, we can use power spectrum density which can correctly reflect the energy level and distribution of the Mu rhythm as another classification feature. Finally, we combine the wavelet coefficient feature represented time domain with the power spectral density represented frequency domain to form a feature vector used for final classification, and get the weight for each feature by iteration in order to get an optimal classification performance.

II. EXPERIMENTAL DATA DESCRIPTION

Experimental data set was obtained from the BCI Competition 2003 provided by Fraunhofer-FIRST, Intelligent Data Analysis Group (Klaus-Robert Müller), and Freie Universität BCerlin, Department of Neurology, Neurophysics Group (Gabriel Curio). The recording was made using a NeuroScan amplifier and an Ag/AgCl electrode cap from ECI 28 EEG channels were measured at positions of the international 10/20-system. This dataset was recorded from a normal subject during a no-feedback session. The subject sat in a normal chair, relaxed arms resting on the table, fingers in the standard typing position at the computer keyboard. The task was to press with the index and little fingers the corresponding keys in a self-chosen order and timing 'self-paced key typing'. The experiment consisted of 3 sessions of 6 minutes each. All sessions were conducted on the same day with some minutes break in between. Typing was done at an average speed of 1 key per second. Dataset are 416 epochs of 500 ms length each ending 130 ms before a key press. Epochs are labeled 0 for upcoming left hand movements and 1 for upcoming right hand movements, and

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316 epochs are for training and the remaining are for test purpose. Data are provided in the original 1000 Hz sampling and 500 samples per channel for each trial [3].

III. PREPROCESSING AND FEATURE EXTRACTION

A. Preprocessing

It is necessary to remove the noise in EEG before further EEG analysis and processing for EEG is deeply masked in the noise background. Our target is to increase signal-to-noise ratio and make it more favorable for feature extraction and pattern recognition.

The frequency of brain electrical activities are mainly in the 0.3-40Hz bands, higher frequency can be seen as noise caused by muscle activity, the blink of eyes and other noise. We use the 5-order band-pass elliptical filtering to do 0.3-40Hz filtering in EEG taking full account to various factors such as signal attenuation during filter and so on. In order to minimize the impact of sampling equipment, we remove 15 points from both sides of the digital signal respectively.

In this research, we choose the signal recorded from electrode C3, C4 and Cz locations of the standard 10/20 system, in primary sensorimotor area of the brain, for EEG pattern recognition, and we only need to preprocess the signal recorded from these three electrodes. Figure 1 shows us the preprocessing course, and the blue line represents the original wave recorded from electrode and the red denoised.

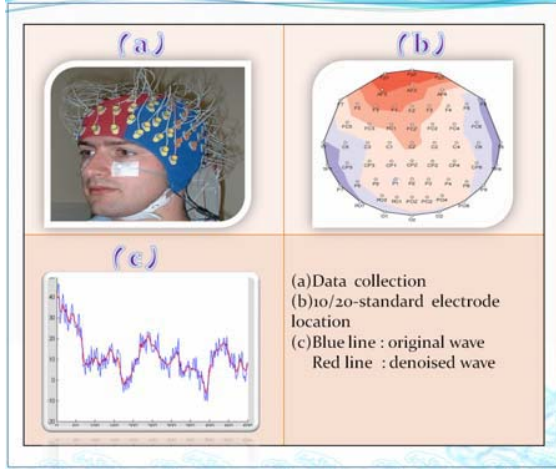


Figure 1. Electrodes locations and C3/Cz/C4 Wave Preprocessing

B. Wavelet Coefficient Feature Extraction

Wavelet transform is a method of multi-resolution time-frequency analysis, which can decompose the mixed signals which consist of different frequencies into different frequency band. EEG signal is analyzed and further denoised using wavelet transform. Moreover, wavelet transform is used for EEG feature extraction in our research. The energies of specific sub-bands and corresponding decomposition coefficients which have maximal separability are selected as features [7]. After trials and statistics, we choose the coefficient $cD6$, $cD7$, $cD8$ well reflecting the

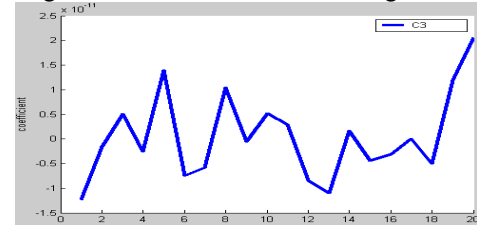
original signal feature as one component of final feature vector.

Before feature extraction, we should eliminate the relevance of signals from different electrodes. Specific way is as formula 1.

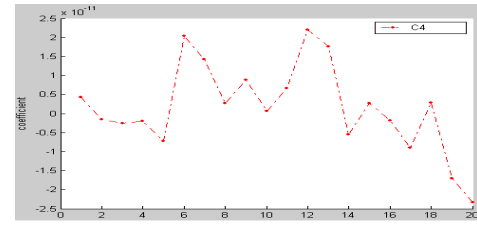
$$l'_m(t) = l_m(t) - \frac{1}{3}(l_{c3}(t) + l_{cz}(t) + l_{c4}(t)) \quad (1)$$

$$m = \{c3, c4, cz\}$$

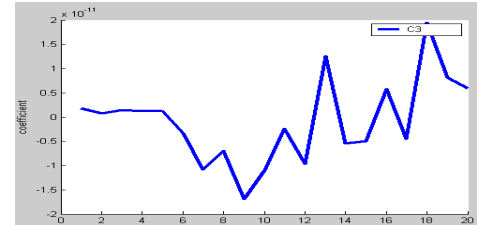
In Formula 1, represented the signal value before eliminating relevance and after eliminating relevance, we



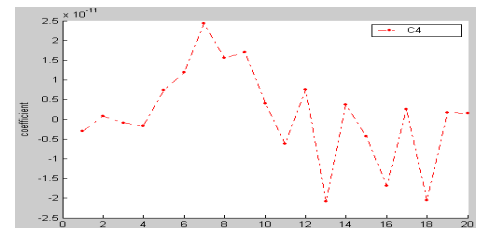
(a) C3/left hand imagery motor



(b) C4/left hand imagery motor



(c) C3/right hand imagery motor



(d) C4/right hand imagery motor

Figure 2. wavelet coefficient in left-right hand imagery movement

can get the coefficient cA5 and cA8 by decomposing the signal in scale 5 and 8, which corresponding to the approximation of original wave in these two level respectively. These coefficients are unexpectedly not same in data length due to differing in decomposition level, so we must reconstruct cA8 in scale 5 and then subtract it from cA5. Finally, we get the wavelet decomposition coefficient cA5-8 which can reconstruct cD6+cD7+cD8 represented the feature of original signal.

Wavelet transform can compress data to a large extent, the compressed data have a dimension of $20 \times 3 \times 316$ compared to the original $500 \times 3 \times 316$.

Figure 2 shows the wavelet coefficient obtained from wavelet decomposition and reconstruction of C3 and C4 in left or right hand imagery motor.

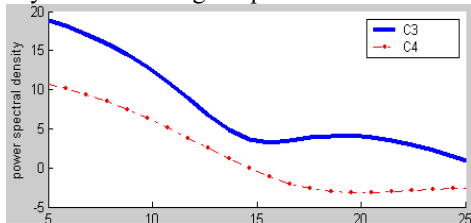
From Figure.2 we can see there's a big discriminant in wavelet coefficient in different imagery motor. It is testified the cubic of coefficient make the discriminant most significant. We can use the difference as one feature in our research. Specific way is as Formula 2.

$$F_c = \sum_{i=1}^{20} C_{c4}^3(i) - \sum_{j=1}^{20} C_{c3}^3(j) \quad (2)$$

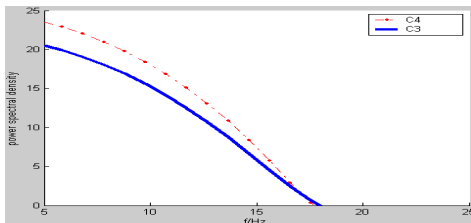
C. Power Spectral Density Feature Extraction

EEG signals recorded are processed using autoregressive (AR) method which is so classical a method in frequency domain [8] and EEG power spectral density is obtained. The parameters of AR method are estimated by the Pwelch method. EEG spectral density is then used to analyze and characterize the left-right imagery movement.

Pwelch, one kind of parametric method, produces better results than classical nonparametric methods by estimating the PSD by first estimating the parameters of the linear



(a) left hand imagery motor



(b) right hand imagery motor

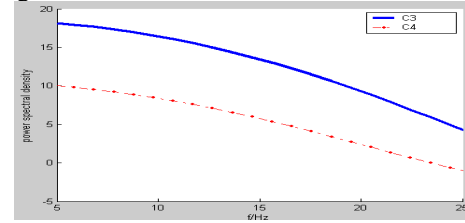
Figure 3. power spectral density with Hamming window 100

system that hypothetically "generates" the signal. But Pwelch depends on a number of parameters such as the type. From Figure 3 we can see the difference between C3 and C4 in the frequency band of Mu rhythm are significant. Finally, we choose the difference as another feature in our research. Specific way is as formula (3), $P(f)$ represents power spectral density.

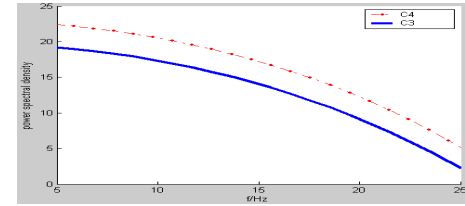
$$F_{P_{xx}} = \sum_{f=8}^{12} P_{xxc3}^2(f) - \sum_{f=8}^{12} P_{xxc4}^2(f) \quad (3)$$

The length of Hamming window is a factor affecting the extent of difference between C3 and C4. By trials and statistics, we can see the length 60 can make better discriminant for most of our training signal.

Figure 4 shows the power spectral density with Hamming window 60.



(a) left hand imagery motor



(b) right hand imagery motor

Figure 4. power spectral density with Hamming window 60

When setting the length of the hamming window 60, more significant difference between C3 and C4 can be see from Figure 4 with comparison of Figure 3.

D. Construction of Feature Vector

We combine the wavelet coefficient feature with the power spectral density as the feature vector F finally fed to the linear discriminant for classification. F is as formula 4.

$$F = \begin{pmatrix} F_{P_{xx}} \\ F_c \end{pmatrix} \quad (4)$$

IV. CLASSIFICATION FOR MOTOR IMAGERY

Classification method has a direct and critical impact on classification performance. There are many ways for classification, such as Linear Discriminant [4], common space models and Bayesian methods and so on. In our research, we choose linear method assigning one weight for each feature of the classification feature vector and

adjusting the weight of different feature through supervised learning. Finally, we can get the optimal weight by iteration which will be used in the determine expression for discriminating left-right hands movement [6].

Specific processes are set out below:

(1) Construct the classification decision function as formula 5.

$$y(F_n) = \alpha F_{p_{xx-n}} + \beta F_{C-n} \quad (5)$$

$$n = 1, 2, \dots, 316$$

In formula 5, the F_n is the feature vector of Nth training trial, $F_{p_{xx-n}}$ and F_{C-n} are components of the feature vector, α and β are random number between 0 and 0.1.

(2) Input the feature vector from training data into formula 5 and adjust α and β according to $y(F_n)$.

a) if $y(F_n) \geq 0$, right hand imagery motor, then $a = a - \Gamma F_{p_{xx-n}}$, $\beta = \beta + \Gamma F_{C-n}$

b) if $y(F_n) < 0$, left hand imagery motor, then $a = a + \Gamma F_{p_{xx-n}}$, $\beta = \beta - \Gamma F_{C-n}$

Γ is a relatively small number between 0 and 1, we choose 0.2 for Γ .

(3) Use step 2 for all group of training data and record the value of α and β every time. Finally, we get a matrix with the dimension of $2 \times 2 \times 316$ for α and β

(4) Get a specific feature decision function with the specific value of α and β , record the classification accuracy with different α and β . Finally, get the best accuracy and α and β in this occasion.

V. CLASSIFICATION RESULT AND DISCUSSION

Classify the test data using the classification decision function we have get. An obtained value $Y(F_n) \geq 0$ and $Y(F_n) < 0$ means that trial n is classified as a left and right trial, respectively.

Figure 5 is the classification result for 100 trials. The average classification accuracy using the feature vector is 82%, higher than that of just using one feature.

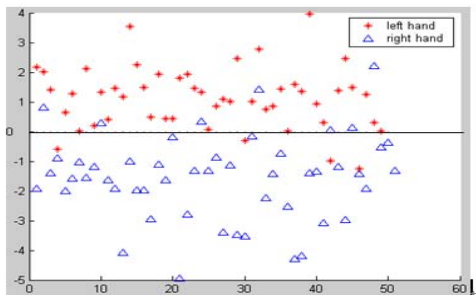


Figure 5. classification result for 100

Our results indicate that the feature vector provides a better classification with accuracy 82% for left-right hands imagery movement compared to the accuracy obtained with only one feature.

There are many reasons for some of the trial results contrary to that of expected, such as the method and parameters for denoising and filtering and the linear classification method is just useful for most of the trails.

VI. CONCLUSIONS AND OUTLOOK

The methods, measurement and classification of the motor imagery signal obtained from specific brain regions, have important theoretical and practical implications for both basic and applied research. The results show that classification accuracy of Motor Imagery EEG was significantly improved. Processing feature vector is a promising algorithm for the analysis of EEG data, deserving to be carefully validated and studied for its own sake. We hope this research has provided new ideas for the design and implementation of Brain-Computer Interface, and be useful in developing a faster and more valuable system adaptive to most users. In addition, it can increase the awareness to this algorithm and to stimulate interest that comparing and contrasting different ways of decomposing EEG data and exploring widely different mental tasks to achieve Brain-Computer Interface [5].

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